

SATYAM KUMAR

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PROFESSIONAL SUMMARY

SDE at Microsoft building database migration infrastructure on Azure — improved PostgreSQL migration success from 54% to 96%, now generalizing the engine for any-to-any database migrations. IIT Kanpur, 4.5 YoE.

WORK EXPERIENCE

Software Development Engineer — Azure PostgreSQL Migration

Feb 2025 – Present

Microsoft IDC

Hyderabad, India

- Architected VM-based dataplane (vmagent + sidecar) for a fault-tolerant, pluggable migration engine, enabling cross-team onboarding onto Azure DMS for migrations to Flexible Server and Horizon DB
- Improved online migration success rate from 54% to 96% by building a pre-migration validation engine and rule framework; integrated Azure Data Explorer into Microsoft Fabric to surface error patterns, driving targeted bug bashes across the migration pipeline
- Replaced wal2json with pgoutput for CDC, resolving OOM on large transactions and achieving 1.8x throughput improvement at 10,000 ops/sec
- Added fault-tolerant retry logic to the migration sidecar with state-aware resumption, eliminating redundant re-processing on failure
- Implemented end-to-end Role, Privilege, and Ownership migration, eliminating the largest pre-migration manual overhead for customers
- Introduced poll-based communication channel from dataplane to control plane via a new finite state machine state and API surface, enabling automated extension allowlisting and locale parameter migration

Associate Software Engineer

Mar 2023 – Jan 2025

A.P. Moller – Maersk

Bangalore, India

- Built four greenfield microservices (Yard, Gate, Dock, Appointment management) in C#/.NET and unified them into an integrated orchestration system, replacing previously siloed manual operations and reducing warehouse labor dependency by ~30%
- Architected multi-tenant RBAC (10+ roles) and site-isolation infrastructure from scratch, scaling to 50+ warehouses each operating as an independent tenant instance

Member of Technical Team

Sep 2021 – Feb 2023

Siply Services Pvt Ltd

Bangalore, India

- Led development of EMI-based gold savings product with reward disbursement (bonus gold + physical delivery), serving 300,000+ users using Node.js and GoLang
- Built event-driven notification and reminder system using AWS SNS/SQS, processing 20,000+ messages/day; adopted across existing products, boosting user engagement by 40%

Software Engineer Intern

May 2020 – Jul 2020

ZestMoney

Remote

- Built real-time event processing system using GoLang, Kafka, and Zookeeper; developed end-to-end KYC onboarding flow in React.js

PROJECTS

URL Shortener | Java, Redis, PostgreSQL | [GitHub](#)

- Built URL shortening service with Redis caching layer for low-latency redirects, PostgreSQL for persistent storage, and collision-resistant key generation

TECHNICAL SKILLS

Languages: Java, Go, C#, JavaScript, SQL

Infra & Cloud: AWS (SQS, SNS, Lambda, EC2), Azure (DMS, VMs), Docker, Kubernetes

Databases: PostgreSQL, Redis, SQL Server, Apache Kafka

EDUCATION

Indian Institute of Technology, Kanpur

Jul 2017 – Jun 2021

Bachelor of Technology — Electrical Engineering | JEE Advanced AIR 1524

Relevant Courses: Machine Learning, Modern Cryptography, Probability & Statistics, Data Structures & Algorithms